

(globals)

Exp8.cpp Exp9.cpp Exp10.cpp Exp11.cpp Exp12.cpp Exp13.cpp [*] Exp14.cpp Exp15.cpp Exp16.cpp Exp17.cpp Exp18.cpp Exp19.cpp

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <windows.h> // FIXED: Using native Windows header instead of pthread.h
4 // Shared variables
5 int counter = 0;
6 CRITICAL_SECTION mutex; // FIXED: Windows equivalent of pthread_mutex_t
7
8 // Function to be executed by threads
9 DWORD WINAPI threadFunction(LPVOID arg) {
10     // Each thread loops 1,000,000 times
11     for (int i = 0; i < 1000000; ++i) {
12         EnterCriticalSection(&mutex); // Lock the critical section
13
14         counter++; // Safely increment the shared variable
15
16         LeaveCriticalSection(&mutex); // Unlock the critical section
17     }
18     return 0;
19 }
20
21 int main(void) {
22     // Initialize the critical section (mutex)
23     InitializeCriticalSection(&mutex);
24
25     HANDLE thread1, thread2;
```

Resources Compile Log Debug Find Results Close

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp19.exe
- Output Size: 129.75390625 KiB
- Compilation Time: 1.92s

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```
25 HANDLE thread1, thread2;
26
27 // FIXED: Moved thread logic inside the actual main function body
28 // Create two separate threads running threadFunction
29 thread1 = CreateThread(NULL, 0, threadFunction, NULL, 0, NULL);
30 thread2 = CreateThread(NULL, 0, threadFunction, NULL, 0, NULL);
31
32 if (thread1 == NULL || thread2 == NULL) {
33     printf("Error creating threads.\n");
34     return 1;
35 }
36
37 // Wait for both threads to completely finish executing
38 WaitForSingleObject(thread1, INFINITE);
39 WaitForSingleObject(thread2, INFINITE);
40
41 // Close thread handles to prevent resource leaks
42 CloseHandle(thread1);
43 CloseHandle(thread2);
44
45 // Destroy the critical section
46 DeleteCriticalSection(&mutex);
47
48 // Print the final value of the counter (Will always be 2000000)
49 printf("Final counter value: %d\n", counter);
```

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(g10Dals)

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49 printf("Final counter value: %d\n", counter);
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51 return 0;
52 }

```

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Compilation

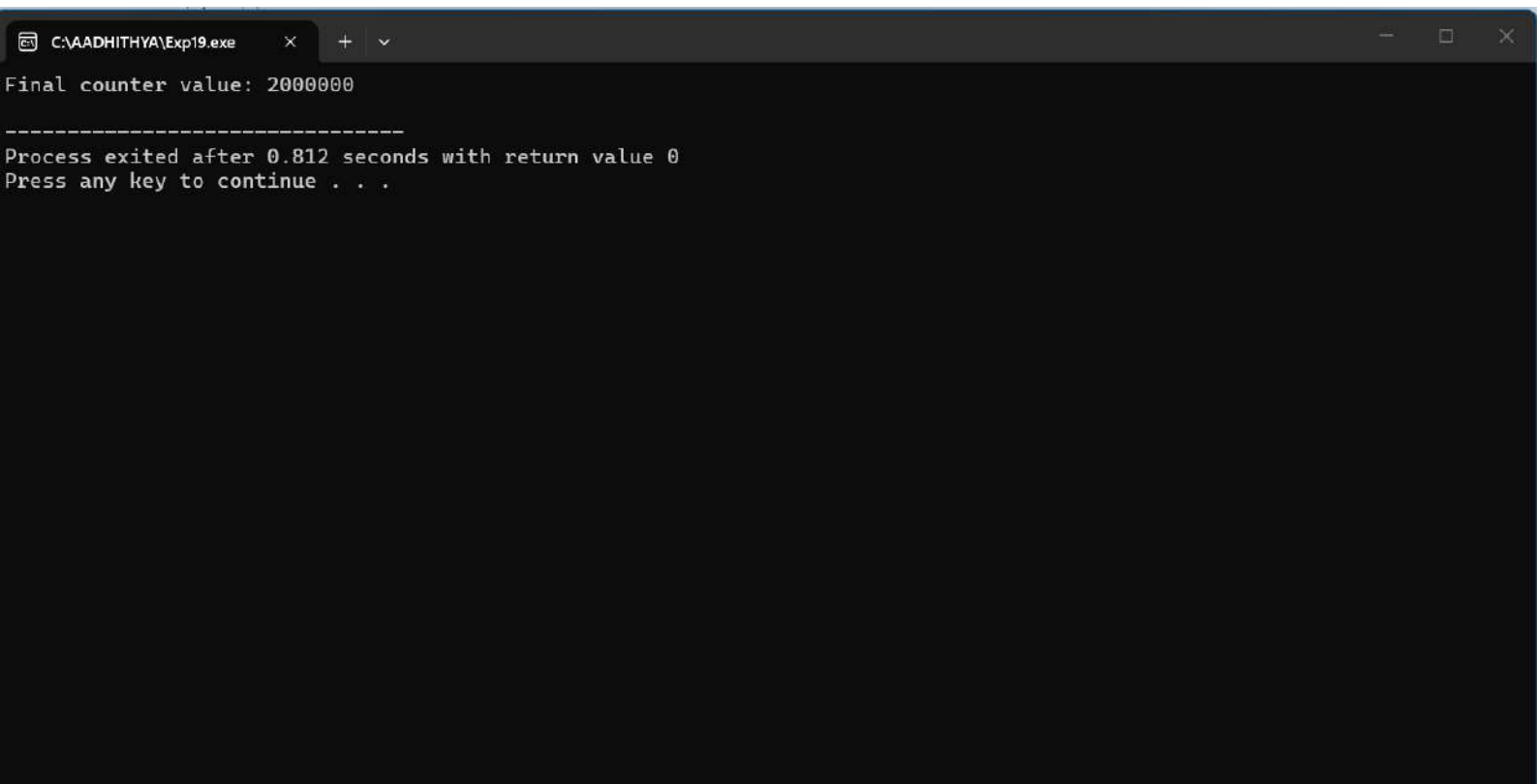
Compilation results...

```

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```

Compiler paths



A screenshot of a Windows command prompt window. The title bar shows the file path "C:\AADHITHYA\Exp19.exe" and standard window controls (minimize, maximize, close). The command prompt displays the following text:

```
Final counter value: 2000000  
-----  
Process exited after 0.812 seconds with return value 0  
Press any key to continue . . .
```